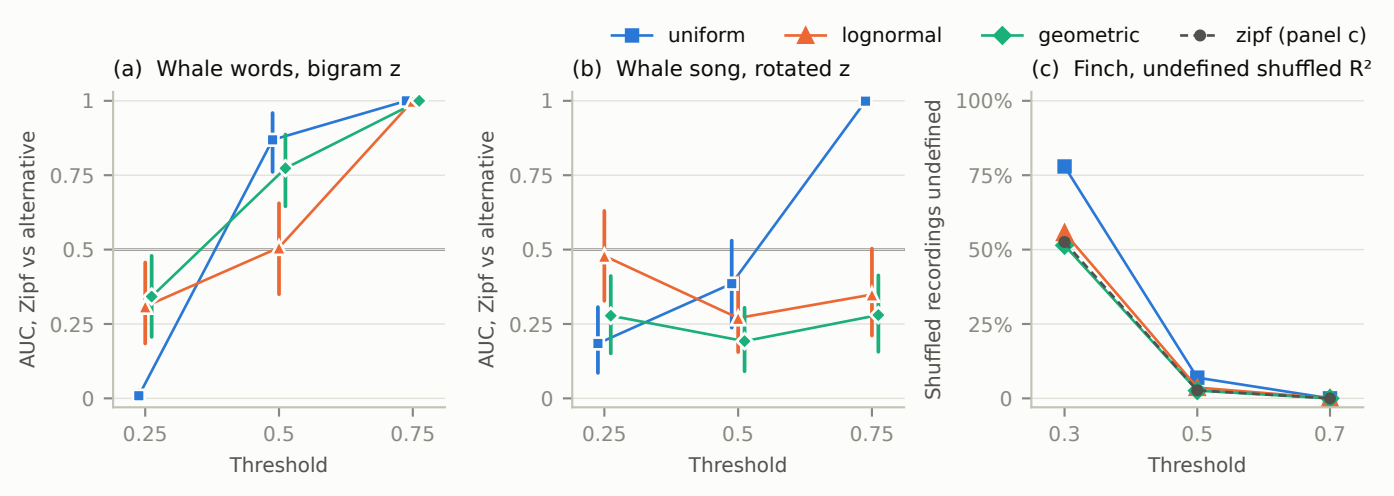

uniform
lognormal
geometric
zipf (panel c)
(a) Whale words, bigram z
(b) Whale song, rotated z
(c) Finch, undefined shuffled R²
AUC, Zipf vs alternative
Shuffled recordings undefined
Threshold
1
0.75
0.5
0.25
0
100%
75%
50%
25%
0.3
0.7